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SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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Technical Presentation

ITA0508-COMPUTER VISION

Technical Analysis of Computer Vision in Healthcare Identifying Real-World Problems, Case Studies, and Professional Insights

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What is Computer Vision in Healthcare?

- Computer Vision enables computers to analyze and interpret medical images.
- Uses Artificial Intelligence (AI) and Deep Learning for diagnosis.
- Supports doctors in detecting diseases accurately and quickly.
- Improves patient care and clinical decision-making.

Real-World Healthcare Problems

- **Problems Addressed by Computer Vision**
- Delayed disease diagnosis.
- Human errors in medical image analysis.
- Increasing workload for radiologists.
- Early detection of cancer and tumors.
- Monitoring patient health remotely.
- Shortage of healthcare professionals.

Applications of Computer Vision

Major Healthcare Applications

- Medical image analysis (X-ray, MRI, CT Scan)
- Cancer detection
- Tumor segmentation
- Diabetic retinopathy detection
- Skin disease identification
- Surgical assistance using robotic vision
- Patient monitoring and rehabilitation

Case Studies & Tools

AI-Based Breast Cancer Detection

- Detects tumors from mammogram images
- Improves early diagnosis and treatment

Diabetic Retinopathy Screening

- Identifies retinal damage from eye images
- Helps prevent vision loss

Common Tools

- OpenCV
- TensorFlow
- PyTorch
- YOLO
- CNN (Convolutional Neural Networks)

Benefits and Challenges

Benefits
- Faster diagnosis
- Higher accuracy
- Early disease detection
- Reduced workload for doctors
- Improved patient outcomes

Challenges
- Large dataset requirements
- Privacy and security concerns
- High implementation cost
- Need for expert validation
- Bias in AI models

Technical Analysis & Insights

- AI models achieve high diagnostic accuracy with quality datasets.
- Image preprocessing improves detection performance.
- Deep learning enables automatic feature extraction.
- Combining Computer Vision with Electronic Health Records (EHR) enhances diagnosis.
- Continuous model training improves reliability.
- Human-AI collaboration provides the best clinical outcomes.

Conclusion

- Computer Vision is transforming modern healthcare.
- It enables faster, more accurate, and reliable disease diagnosis.
- Despite challenges, ongoing advancements improve clinical performance.
- Future developments will make healthcare more efficient, accessible, and patient-centered.

Thank You